

Nibrexo OS AI — Technical Architecture Document

v1.0

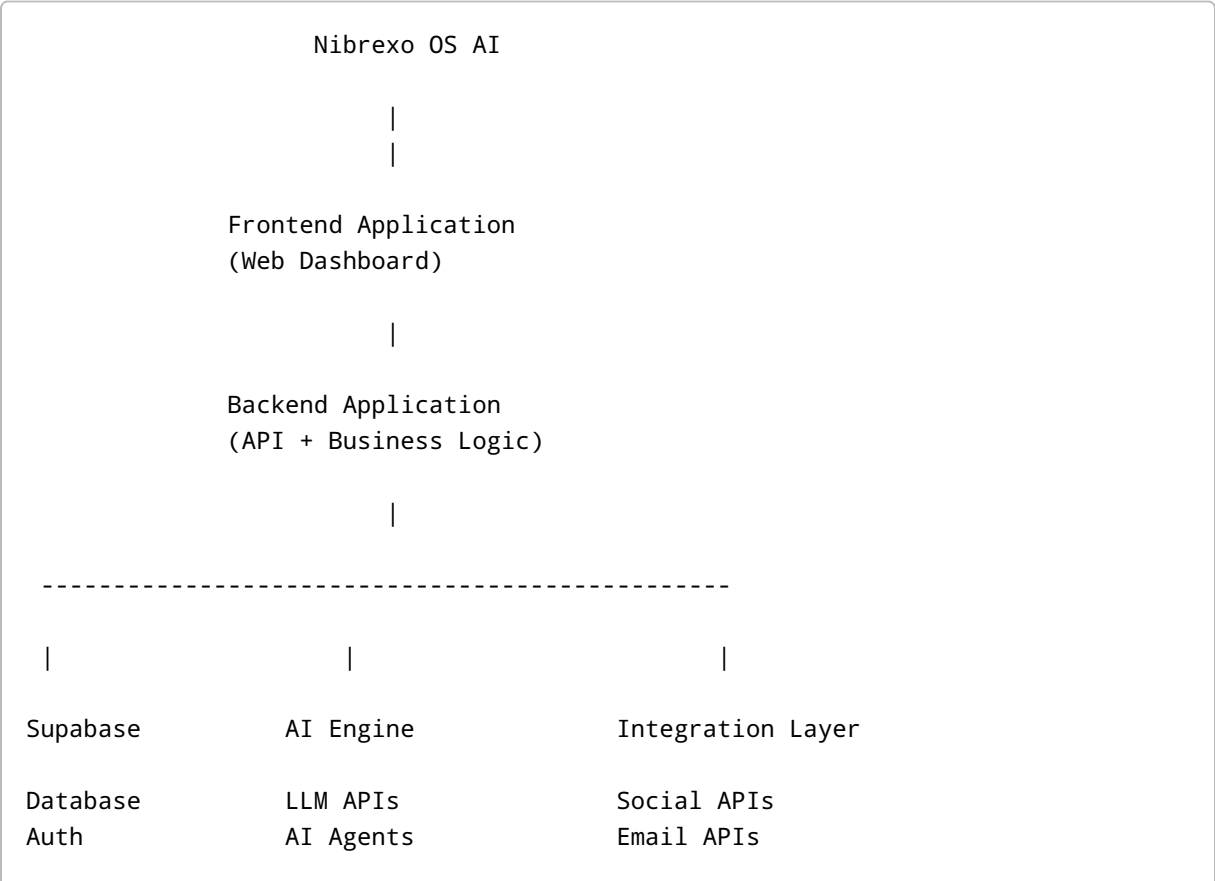
1. Architecture Overview

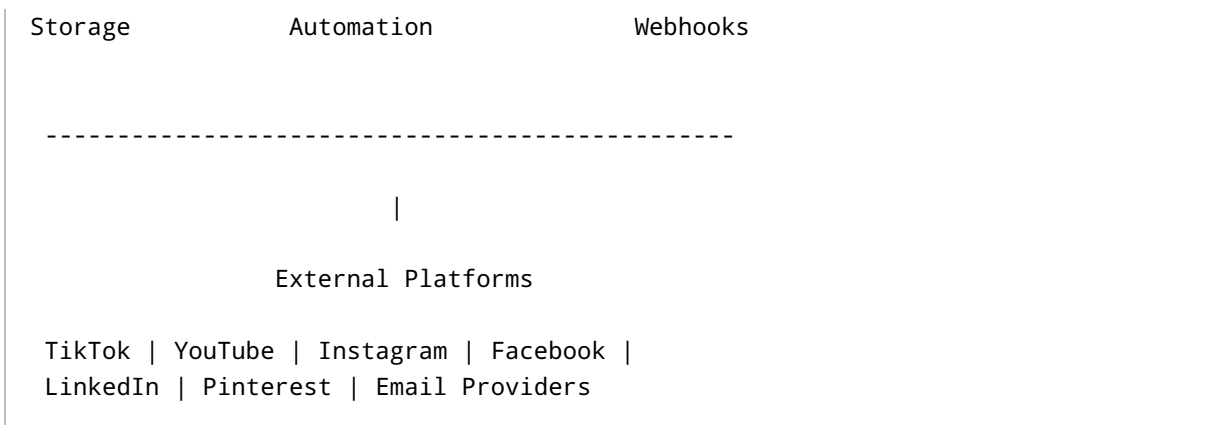
Nibrexo OS AI ek scalable AI-powered SaaS platform hoga jo frontend, backend, database, AI services aur external platform integrations ko ek secure architecture ke through connect karega.

Primary goals:

- High scalability
- Secure data handling
- Easy maintenance
- Future feature expansion
- Complete ownership by Nibrexo

2. High-Level System Architecture





3. Frontend Architecture

Technology Stack

Recommended:

- Next.js
- React
- TypeScript
- Tailwind CSS

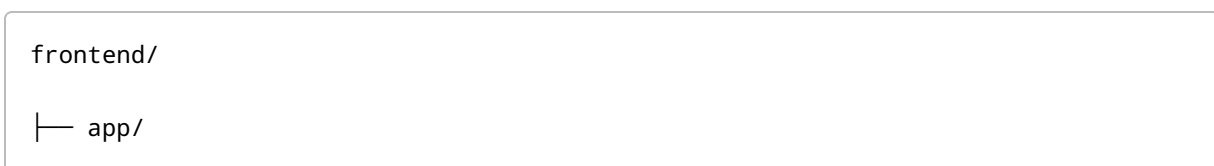
Purpose

Frontend Nibrexo users ko ek professional dashboard provide karega jahan se woh:

- Social accounts manage
- Messages view
- Content upload
- Emails create
- AI assistant use
- Analytics check

kar sakege.

4. Frontend Structure



```
|
|— components/
|
|— dashboard/
|
|— social/
|
|— inbox/
|
|— email/
|
|— ai-assistant/
|
|— settings/
|
|— auth/
|
|— utils/
```

5. Backend Architecture

Purpose

Backend Nibrexo OS AI ka main control layer hoga.

Responsibilities:

- Authentication handling
- API communication
- Business rules
- AI requests
- Platform connections
- Automation workflows

Backend Structure

```
backend/
|
|— api/
|
|— services/
```

```
|   ├── social-service
|   ├── email-service
|   ├── ai-service
|   └── user-service
```

```
└── middleware/
```

```
└── database/
```

```
└── security/
```

```
└── config/
```

6. Database Architecture (Supabase)

Supabase Nibrexo OS AI ke liye:

- Database
- Authentication
- File storage
- Security policies

handle karega.

Main database areas:

Users

Store:

- Account information
- Roles
- Permissions

Social Accounts

Store:

- Connected platforms
- Account details

- Secure tokens
 - Connection status
-

Content

Store:

- Videos
 - Images
 - Captions
 - Scheduling data
 - Publishing status
-

Messages

Store:

- Platform messages
 - Comments
 - Replies
 - Conversation history
-

Clients

Store:

- Client details
 - Communication history
 - Project information
-

Email System

Store:

- Templates
 - Campaigns
 - Sending history
 - Follow-ups
-

7. AI Layer Architecture

AI layer Nibrexo OS AI ka intelligence center hoga.

Functions:

Content AI Agent

Responsibilities:

- Captions generate karna
 - Video descriptions
 - Hashtags
 - Content ideas
-

Communication AI Agent

Responsibilities:

- Messages analyze karna
 - Reply suggestions
 - Customer intent identify karna
 - Brand voice maintain karna
-

Email AI Agent

Responsibilities:

- Professional emails create karna
 - Templates customize karna
 - Follow-up sequences generate karna
-

8. Social Integration Layer

Integration layer external platforms ko manage karegi.

Supported:

- YouTube API
- Meta APIs

- TikTok API
- LinkedIn API
- Pinterest API

Responsibilities:

- Authentication
- Upload requests
- Data synchronization
- Webhook handling

9. Email System Architecture

Email automation ke liye external email delivery provider APIs use hongy.

Flow:

User Request

↓

AI Email Generator

↓

Template Selection

↓

Approval

↓

Email Provider API

↓

Client Email Delivery

10. Authentication & Security

Security requirements:

- Secure login
- Role-based permissions
- Encrypted API credentials
- Protected environment variables
- Database access policies

Sensitive information:

- API keys
- Access tokens
- Client data

kabhi frontend mein expose nahi hogi.

11. Hosting & Deployment Architecture

Development Flow

Developer

↓

GitHub Repository

↓

Version Control

↓

Vercel Deployment

↓

Production Environment

↓

Custom Domain

12. GitHub Repository Strategy

Recommended:

```
nibrexo-os-ai/  
├─ frontend  
├─ backend  
├─ database  
├─ documentation  
├─ tests  
└─ README
```

Branches:

```
main  
|  
development  
|  
feature branches
```

13. Environment Management

Example:

```
Production Environment  
  
SUPABASE_URL  
SUPABASE_KEY  
AI_API_KEY  
SOCIAL_API_KEYS  
EMAIL_API_KEYS
```

Rules:

- Secrets GitHub par upload nahi hong
 - Environment variables use hong
 - Access limited hoga
-

14. Future Scalability Plan

Architecture future mein support kare:

- Multiple companies
 - Multiple team members
 - Client workspaces
 - Mobile applications
 - Advanced AI agents
 - Analytics system
 - Marketplace features
-

Document Status

Version: 1.0

Module: Technical Architecture

Product: Nibrexo OS AI

Owner: Nibrexo

Next Section:

Complete Feature Requirements + User Roles & Permissions